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Assessment of Post-monsoon Drought Over Marathawada Region (Maharashtra, India) Using MODIS Data

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Abstract Drought is a natural hazard that has a significant effect on the socio-economic, agricultural, and environmental aspects of a region. The Marathwada division of Maharashtra state is infamous for recurring drought situations. Poor precipitation, lack of water storage, relatively high temperature in pre and post-monsoon seasons, and variable unfavorable weather conditions lead to drought in this region. Post-monsoon drought mainly occurs due to deficit rainfall and a sudden increase in temperature. This study represents the overall assessment of post-monsoon drought over Marathwada during October, November, and December for the period 2001 to 2017. Monsoon rainfall deficit results in post-monsoon drought in subsequent months. Detection and monitoring of drought over large areas are possible through remote sensing indices namely, Temperature Condition Index (TCI), Vegetation Condition Index (VCI), and Vegetation Health Index (VHI). VHI is the resultant index from temperature and vegetation indices, which helps to understand vegetation health. During the last 17 years, moderate to severe drought has been observed in two successive years of 2014 and 2015, where TCI, VCI, and VHI indices indicated these years as drought-prone years for the post-monsoon season. Particularly, Bid, Osmanabad, Latur, Nanded, and Parbhani districts suffered severe drought in these successive years. Whereas, all the other years except 2010 and 2017 experienced normal conditions to moderate drought in the Marathwada region. Due to erratic rainfall, it is necessary to plan water utilization and storage in Marathwada to overcome the recurrent drought experienced in the region. This may help agronomists and planners in better management of water resources, particularly for the agricultural sector.

Keywords: *drought, TCI, VCI, VHI, Marathwada, MODIS*

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1. Introduction

Drought is a natural hazard occurring mainly due to lack or deficiency of precipitation and adversely affecting all human activities of the area, primarily agriculture. Its onset and evolution are very slow and therefore it is difficult to predict. It is often said that drought is a most complex natural hazard than any other, and it affects a large number of people [1]. It has significant adverse effects on the economy, agriculture, and environmental conditions. High evapotranspiration, scanty rainfall and overuse or exploitation of water resources, or a combination of these parameters and insufficient moisture content in the atmosphere might be responsible for drought conditions [2,3]. Direct impacts of drought are forest fires, reduced crop production, and water level, damage of fish habitat, livestock mortality rate, and its indirect effects include reduction in crop production, the increased market value of goods, vegetables, and other commodities [4].

In India, several studies have been done on drought monitoring using spatial indices on various geographic regions like western India [4] Aravalli Region [2], Indian Gangetic Plains [6], Gujarat [6], and Maharashtra [8]. It is important to monitor drought continuously in time and space during the drought period with the help of Palmer Drought Severity Index (PDSI), Standardized Precipitation Index (SPI), and other indices [9]. In India, Rajasthan, Kutch of Gujrat, Marathwada, and Vidarbha in Maharashtra, and some parts of Orrisa are major drought-prone areas (DPA). Marathwada is one of the DPA in Maharashtra which has serious issues related to agricultural crops, water storage, and temperature increase since the last decade.

Various studies in the past have used geospatial data and indices derived from them to study drought. Some studies have used NOAA AVHRR data to estimate

temperature condition index (TCI) and vegetation condition index (VCI) [10,11] which were ultimately used to derived vegetation temperature condition index (VTCI) using the formula VTCI = 0.7×VCI+0.3×TCI. A comparative study has been carried out between SPI, Standardized Water-level Index (SWI), and VHI for the pre-monsoon and monsoon season over the Aravalli part of India [2]. According to this study, meteorological and hydrological conditions have to be considered to understand the phenomenon of drought. Droughts are estimated by using TCI, VCI and VHI validated through SPI and SWI [6]. Landsat data has been used for estimation of TCI, VCI, and VHI where Digital Numbers (DN) values, Brightness Temperature (BT), Normalized Difference Vegetation Index (NDVI), and Land Surface Temperature (LST) algorithm has been used to estimate drought over Lebanon [2]. MODIS instrument is another way to achieve RS data for EVI and LST. Indices derived were TCI, VCI, and VHI and compared with each other over East Java during El Nino year of 2015 [2]. Different indices discussed above reveal different information relevant to drought studies.

It has been reported that extremely hot days during winter and summer are increasing in some parts of Maharashtra and Karnataka [2] which makes it imperative to analyze drought conditions over this area. Also, in Maharashtra, over Vidarbha and Marathwada, scarcity of rainfall and misuse of available groundwater leads to droughts in almost every successive year and an increasing rate of farmer suicides is one of the consequences of increasing droughts in this region. Changing agricultural practices, less rainfall, water scarcity, and changing weather conditions play a crucial role in incidences of droughts in the study region.

2. Study Area and Data Used

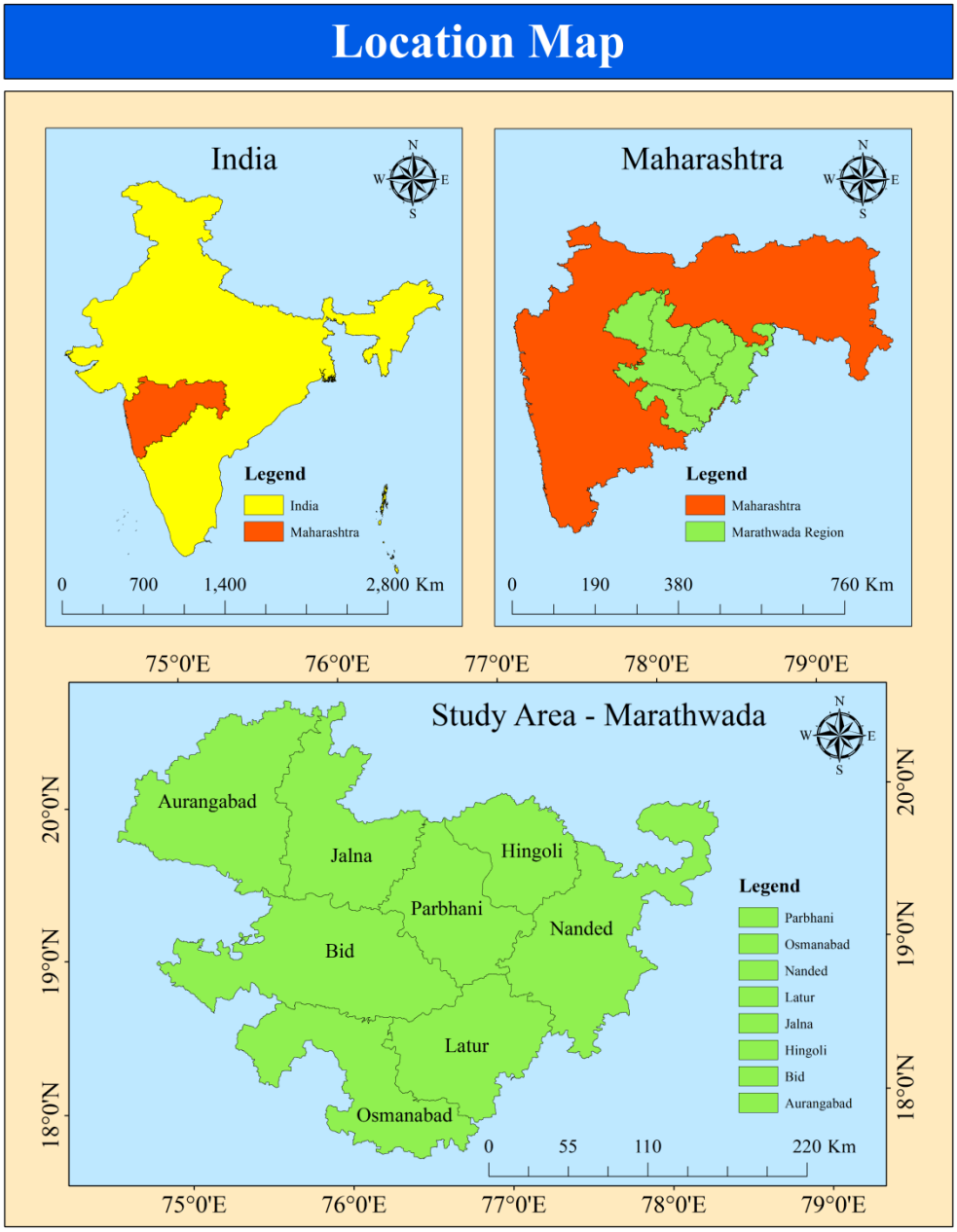


Figure 1. Location map

The main aim of the study is to investigate spatial and temporal aspects of drought over the Marathwada region (Figure 1) of Maharashtra (India) during the post-monsoon season using various indices like TCI, VCI, and VHI for the period of 17 years from 2001 to 2017. Marathwada region includes 8 districts of Maharashtra namely, Aurangabad, Nanded, Latur, Parbhani, Jalna, Beed, Hingoli, and Osmanabad. The total geographical area of the region is 64590 km^2. The major part of the region is located in the Godavari river basin, the largest river in southern India. The study area is one of the low precipitation zones in Maharashtra with an annual rainfall of approximately 882 mm from June to September (Source: http://www.rainwaterharvesting.org/urban/rainfall.htm). In the last 15 years of the study period, the lowest annual rainfall is observed over Aurangabad and the highest rainfall is received over Nanded. The study has been carried out in the post-monsoon season and it is known as "Rabi Crop Season" in Maharashtra. The average day temperature (maximum temperature) ranges from 28°C to 38°C and the average night temperature (minimum temperature) ranges from 20°C to 27°C. During summer temperature goes around 46°C in Nanded.

Satellite data and remote sensing technology play a vital role in monitoring natural hazards as well as crop health and development related to climatic conditions. The present study has utilized satellite data from Moderate Resolution Imaging Spectro Radiometer (MODIS) on-board terra satellite with two products namely MOD11A2 and MOD13A2. MOD1A2 is LST 8 days' composite L3 product at 1 km spatial resolution with emissivity bands whereas, MOD13A2 is NDVI and EVI 16 days' composite L3 product at 1 km spatial resolution available in the sinusoidal grid.

3. Pre-processing

The present study is based on Spatio-temporal analysis along with trend analysis of temperature and vegetation condition. All the satellite data has been projected to Albers Conical Equal Area projection with WGS1984 Datum. For the study, ENVI, ArcGIS, and ERDAS Imagine softwares has been used. For statistical analysis, ArcGIS and Microsoft Excel have been used. The study is carried out over Marathwada for the period of 17 years (2001 to 2017) and therefore required satellite data were acquired for 17 years.

4. Methodology

All pre-processed data of MOD11A2 and MOD13A2 is used to estimate TCI, VCI, and VHI using the methodology illustrated in Figure 2. Minimum and Maximum LST and NDVI are required to run the algorithm for achieving TCI and VCI, which were derived by building a model using the ERDAS Imagine software. The vegetation health index was computed using TCI and VCI index.

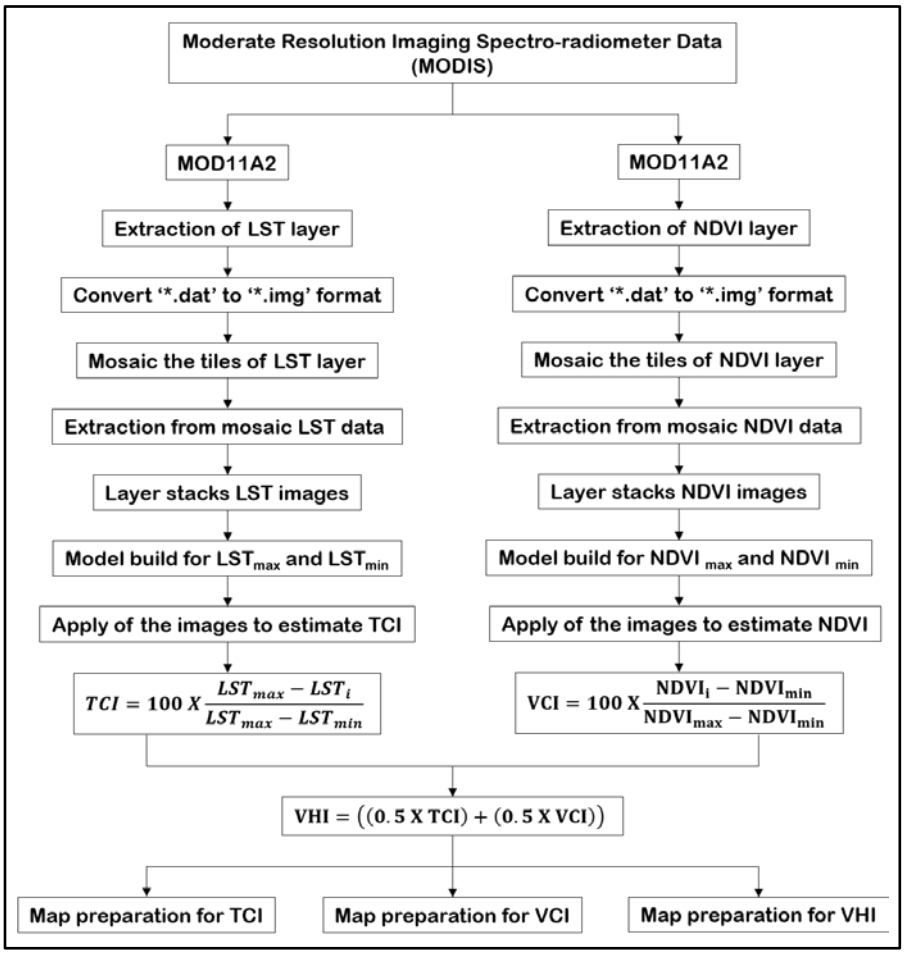


Figure 2. Methodology Tree Diagram

4.1. Remote Sensing Indices

There are several types of droughts among them agricultural drought has been monitored using the most efficient and popular spectral indices such as TCI, VCI, and VHI, which are focused on temperature-based stress, vegetation stress, and overall health of the vegetation respectively.

4.1.1. Temperature Condition Index (TCI)

Temperature Condition Index is used to estimate vegetation stress due to temperature and excessive wetness [10]. It represents relative changes in the thermal conditions obtained from land surface temperature data. Changes in the vegetation health due to thermal stress can be observed using TCI. It can be expressed in the formula as,

$$TCI = 100X \frac{LST_{\max} - LST_i}{LST_{\max} - LST_{\min}} \quad (1)$$

Where, LST_i, LST_{min}, and LST_{max} are defined as LST of the current month, minimum, and maximum LST values in multi-year series respectively. TCI values range from 0 to 100, where low TCI indicates unfavorable condition with high temperature and dryness and high TCI values indicates favorable condition with minimal temperature stress. Values near and more than 50 indicate moderate to good vegetation condition due to temperature.

4.1.2. Vegetation Condition Index (VCI)

The Vegetation condition index is a NDVI normalization pixel-based index, where it shows vegetation health due to vegetation condition. Stressed vegetation with lower NDVI depicts poor health of vegetation whereas NDVI of 1 depicts dense vegetation with good health. The use of NDVI is the primary tool for the description of vegetation phenology, continental land cover, vegetation classification, and dynamics and cropping practices [2]. VCI can be expressed as:

$$VCI = 100X \frac{NDVI_i - NDVI_{\min}}{NDVI_{\max} - NDVI_{\min}} \quad (2)$$

Where $NDVI_i$, $NDVI_{min}$, and $NDVI_{max}$ are defined as NDVI of the current month, minimum, and maximum of NDVI values in multi-year series respectively. VCI values range from 0 to 100, where low VCI values indicate unfavorable condition with high dryness and high VCI values indicates the favorable condition. This index indicates current vegetation conditions. VCI values around 50 suggest fair vegetation condition and values between 50 and 100 suggest good condition.

4.1.3. Vegetation Health Index (VHI)

This index is the combination of the moisture and thermal condition of vegetation which shows overall vegetation health [14]. Equal weights have been assigned to TCI as well as VCI since the moisture and temperature condition during the vegetation cycle is currently not known and it is assumed that the share of weekly TCI and VCI is equal [14]. Extremely unhealthy vegetation conditions (low VHI) are normally associated with both severe moisture stress (VCI) and thermal stress (TCI) and vice versa [14].

Vegetation Heath Index can be expressed as:

$$VHI = \left((0.5X\ TCI) + (0.5X\ VCI)\right) \quad (3)$$

Lower values of VHI have a greater intensity of drought whereas higher values of VHI have lower intensity of drought. VHI has been used for many applications depending on purposes, like drought severity, time and period of drought, and normal drought identification. There are 5 classes of drought for identifying the vegetation health or drought condition.

Table 1. Drought class for VHI

Drought Class	VHI
Extreme Drought	< 10
Severe Drought	≥ 10 and ≤ 20
Moderate Drought	≥ 20 and ≤ 30
Mild Drought	≥ 30 and ≤ 40
No Drought	≥ 40

5. Results and Discussion

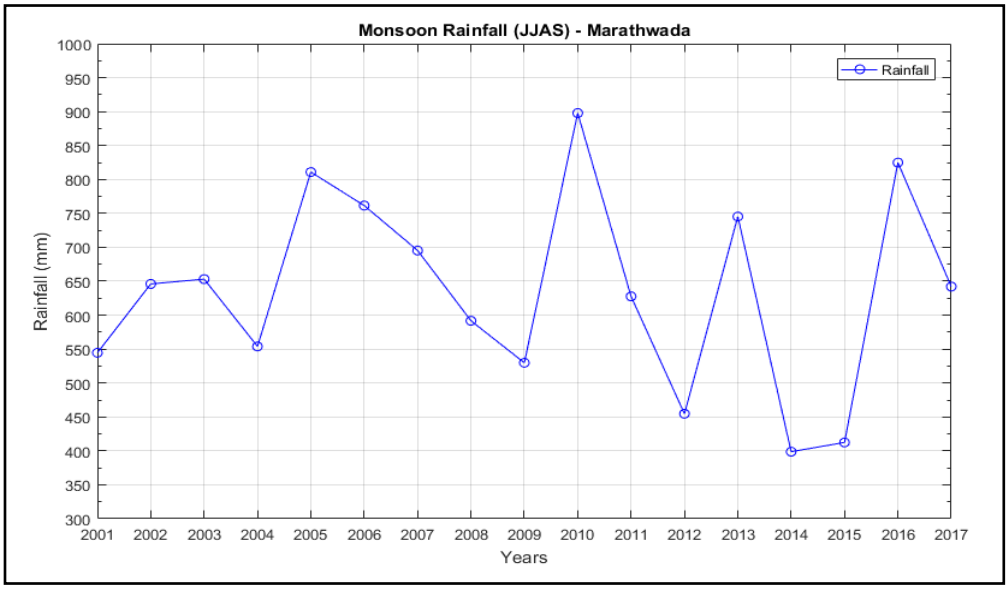


Figure 3. Rainfall during Monsoon (JJAS) – Marathwada

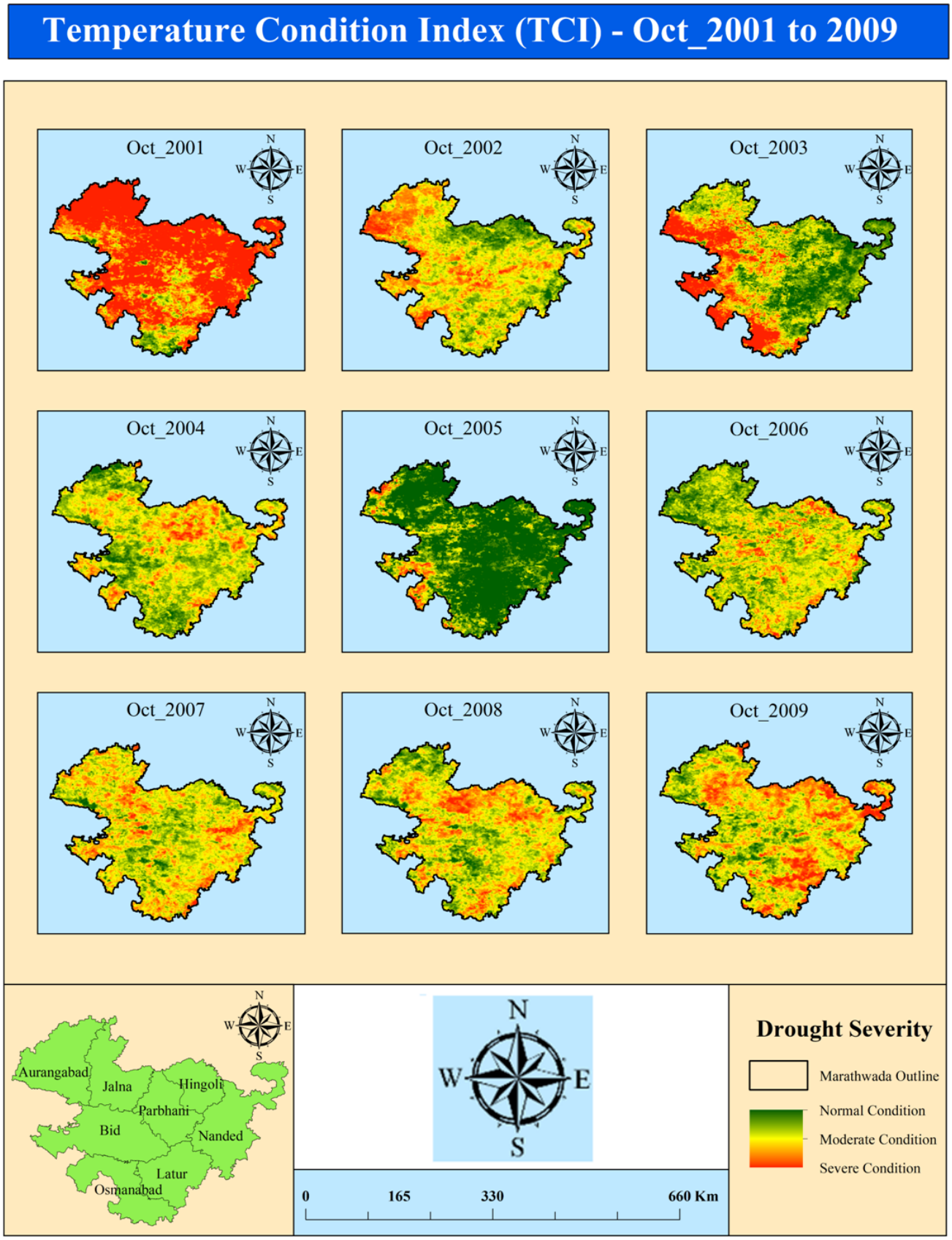


Figure 4. Temperature Condition Index (TCI) - October 2001 to 2009

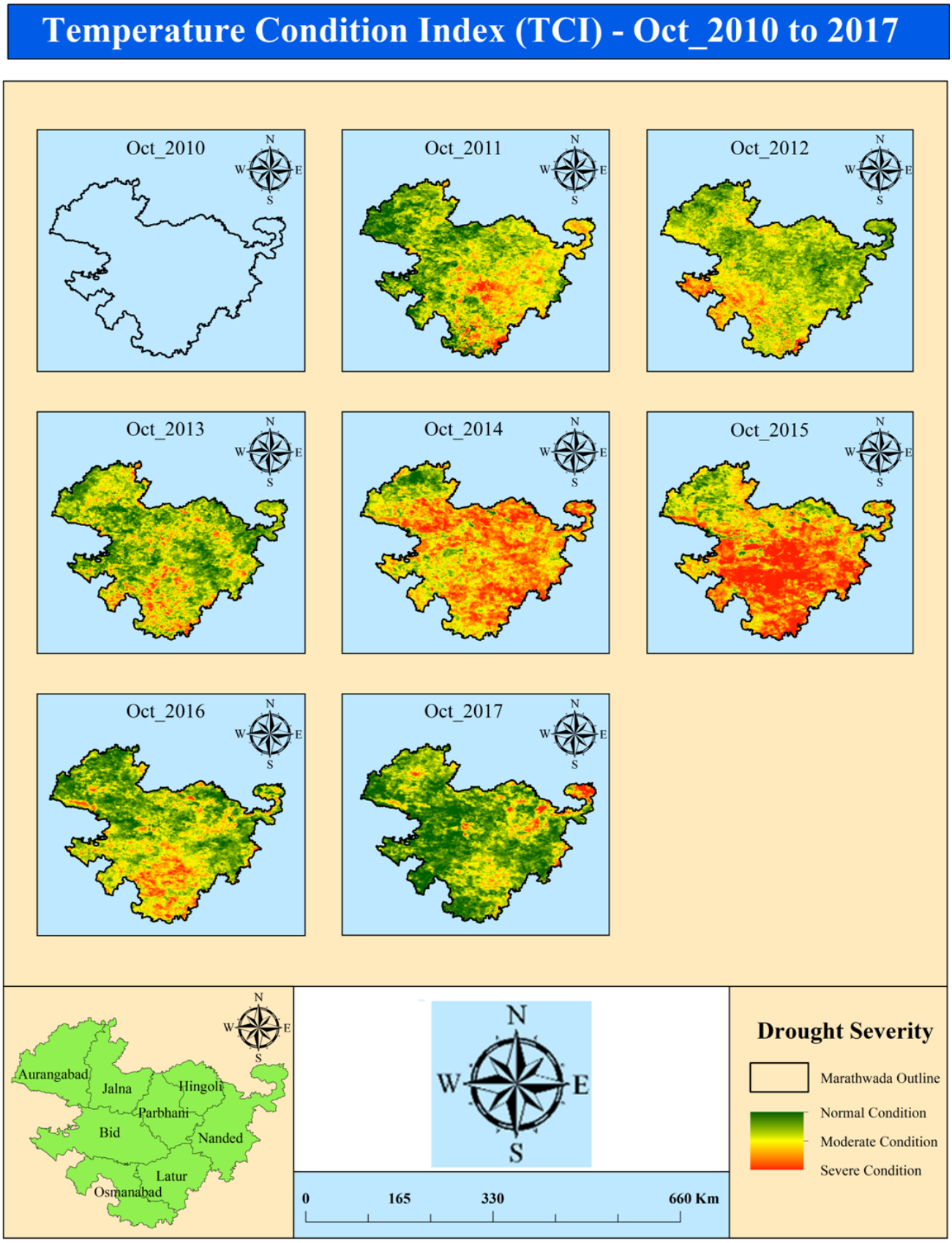


Figure 5. Temperature Condition Index (TCI) - October 2010 to 2017

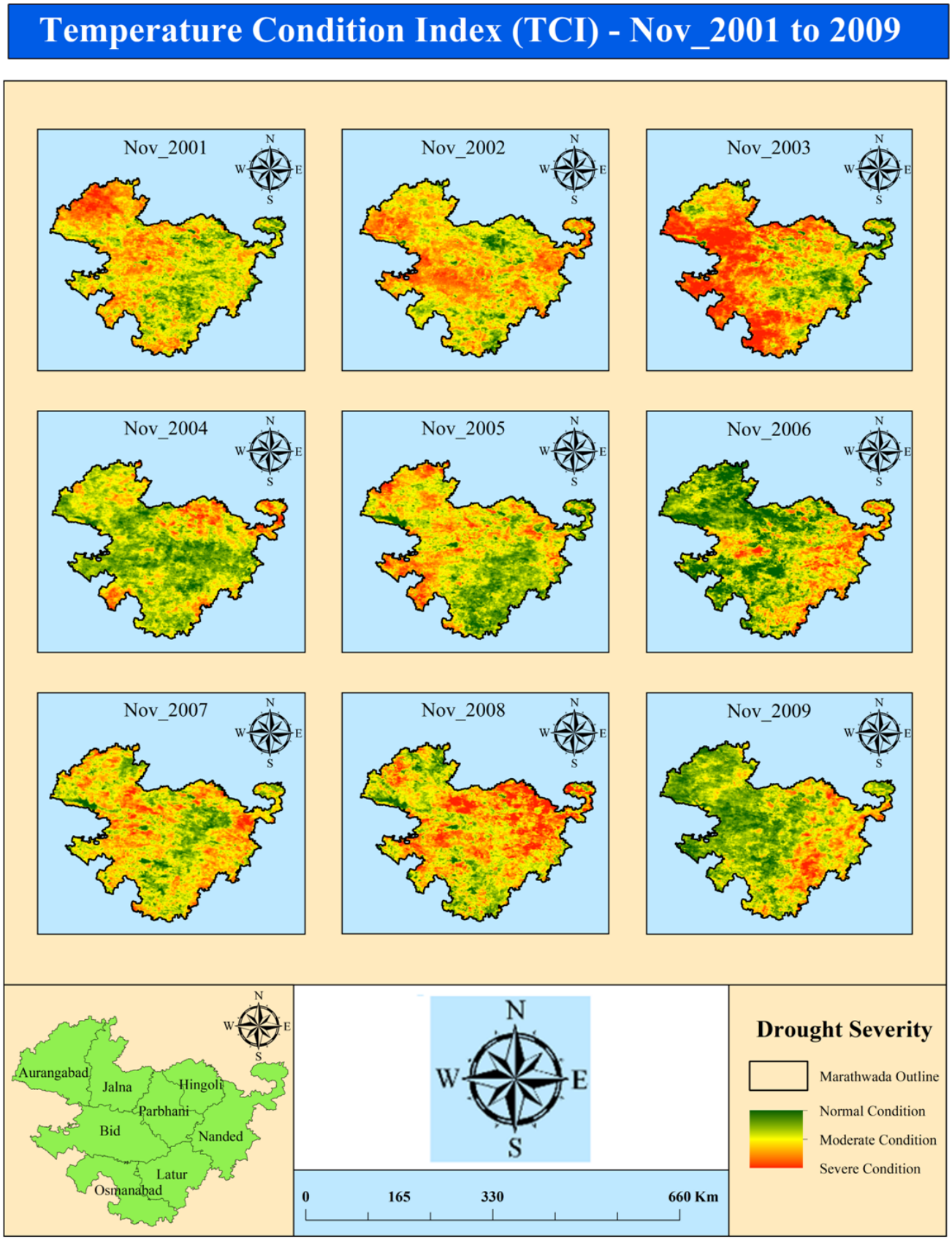


Figure 6. Temperature Condition Index (TCI) - November 2001 to 2009

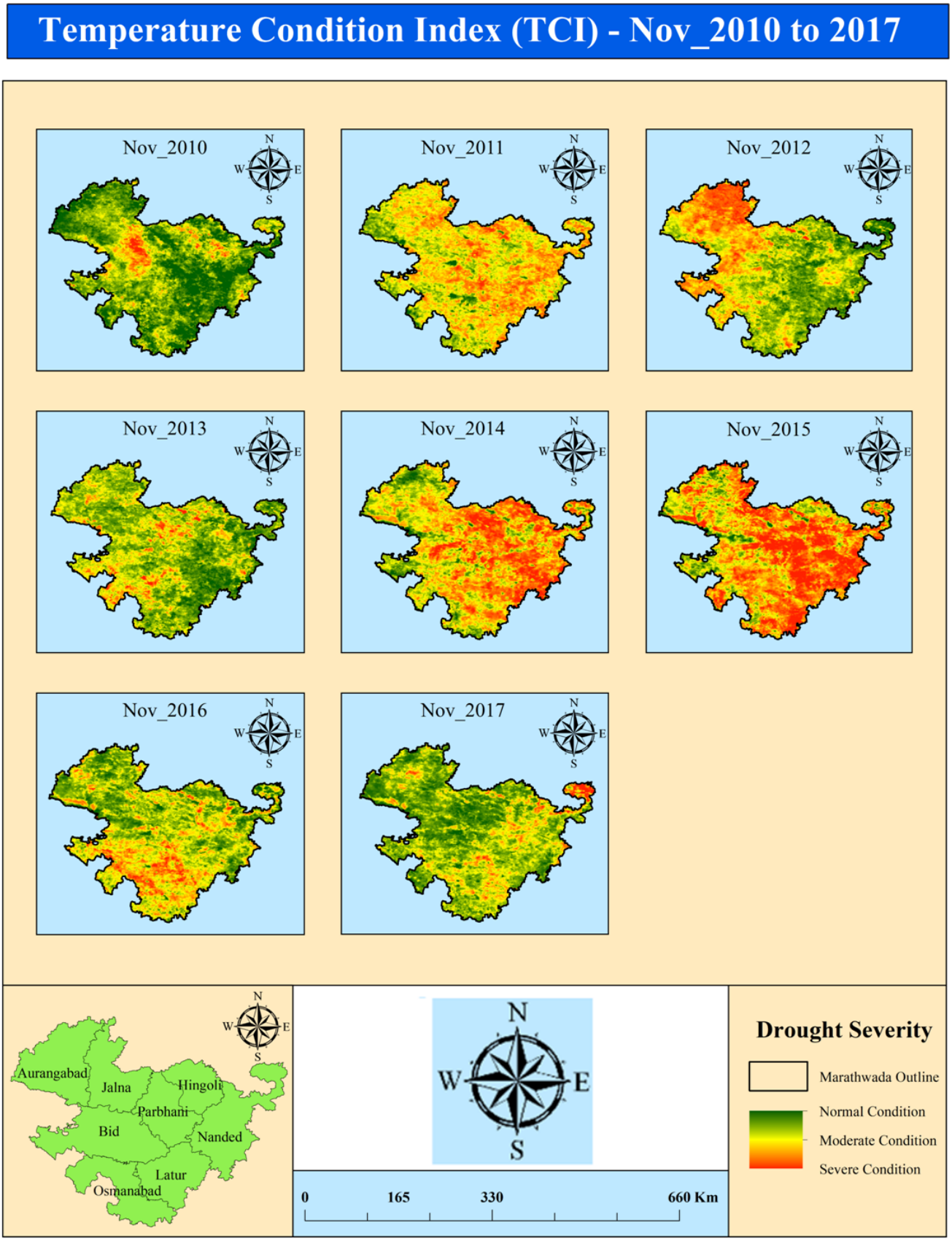


Figure 7. Temperature Condition Index (TCI) - November 2010 to 2017

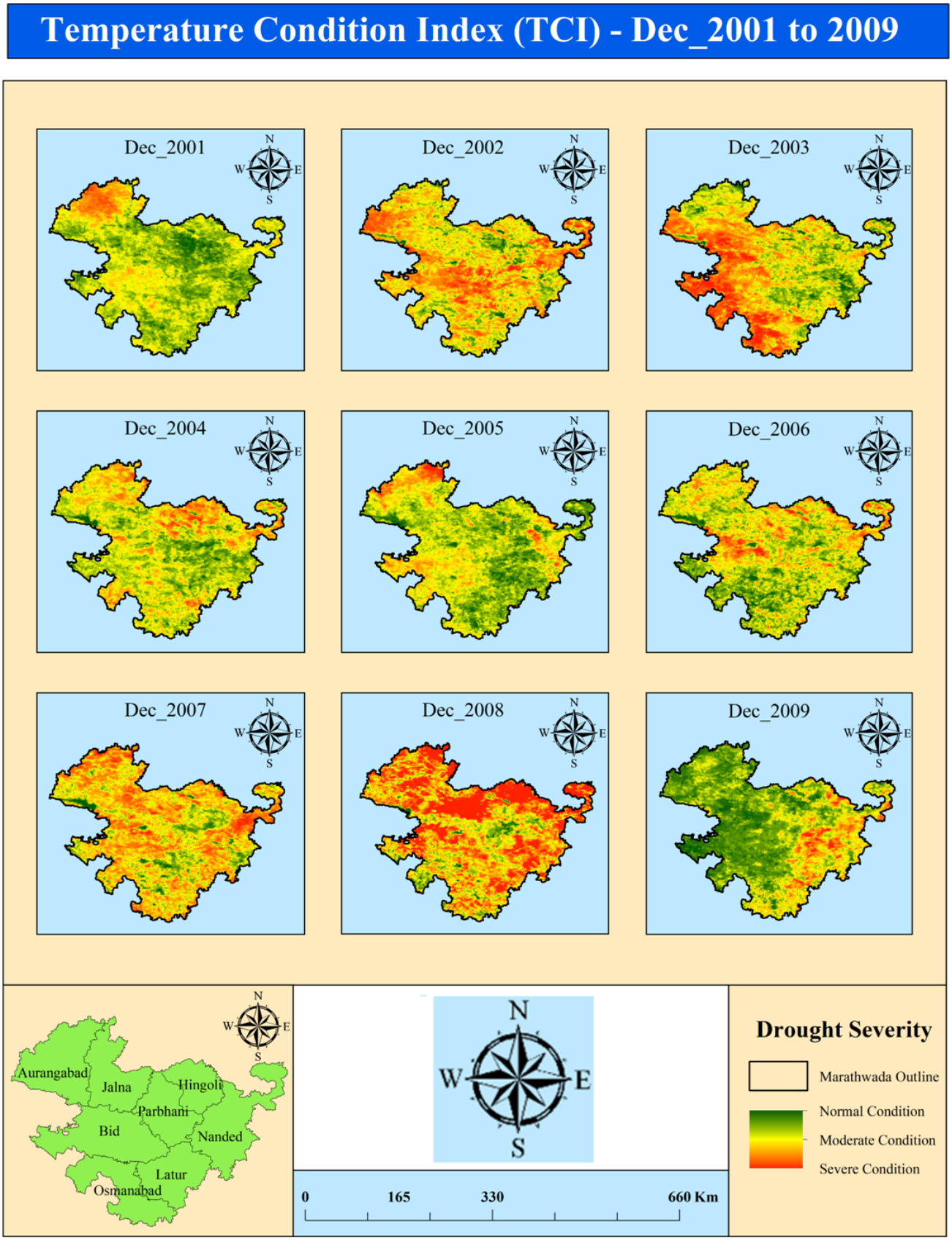


Figure 8. Temperature Condition Index (TCI) - December 2001 to 2009

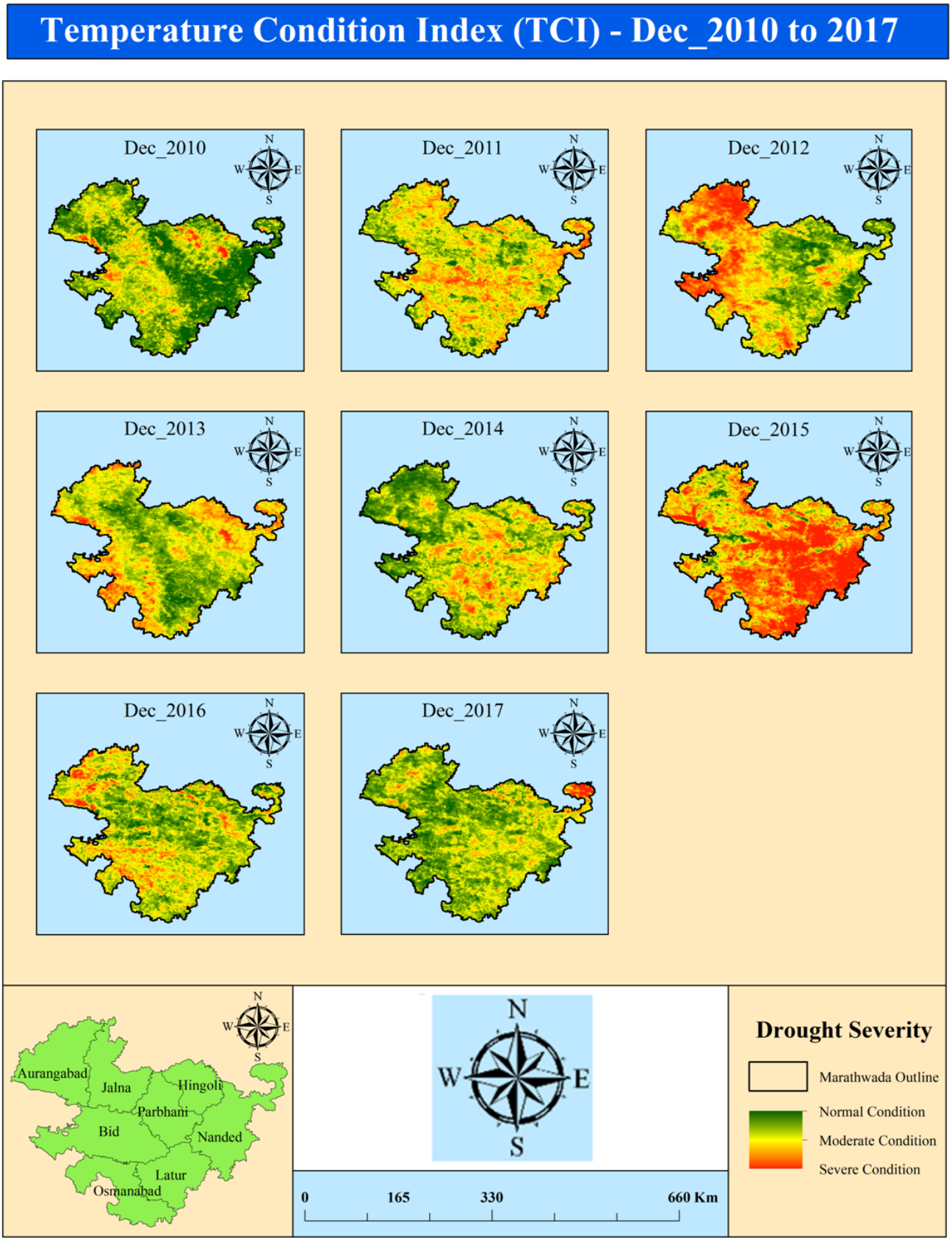


Figure 9. Temperature Condition Index (TCI) - December 2010 to 2017

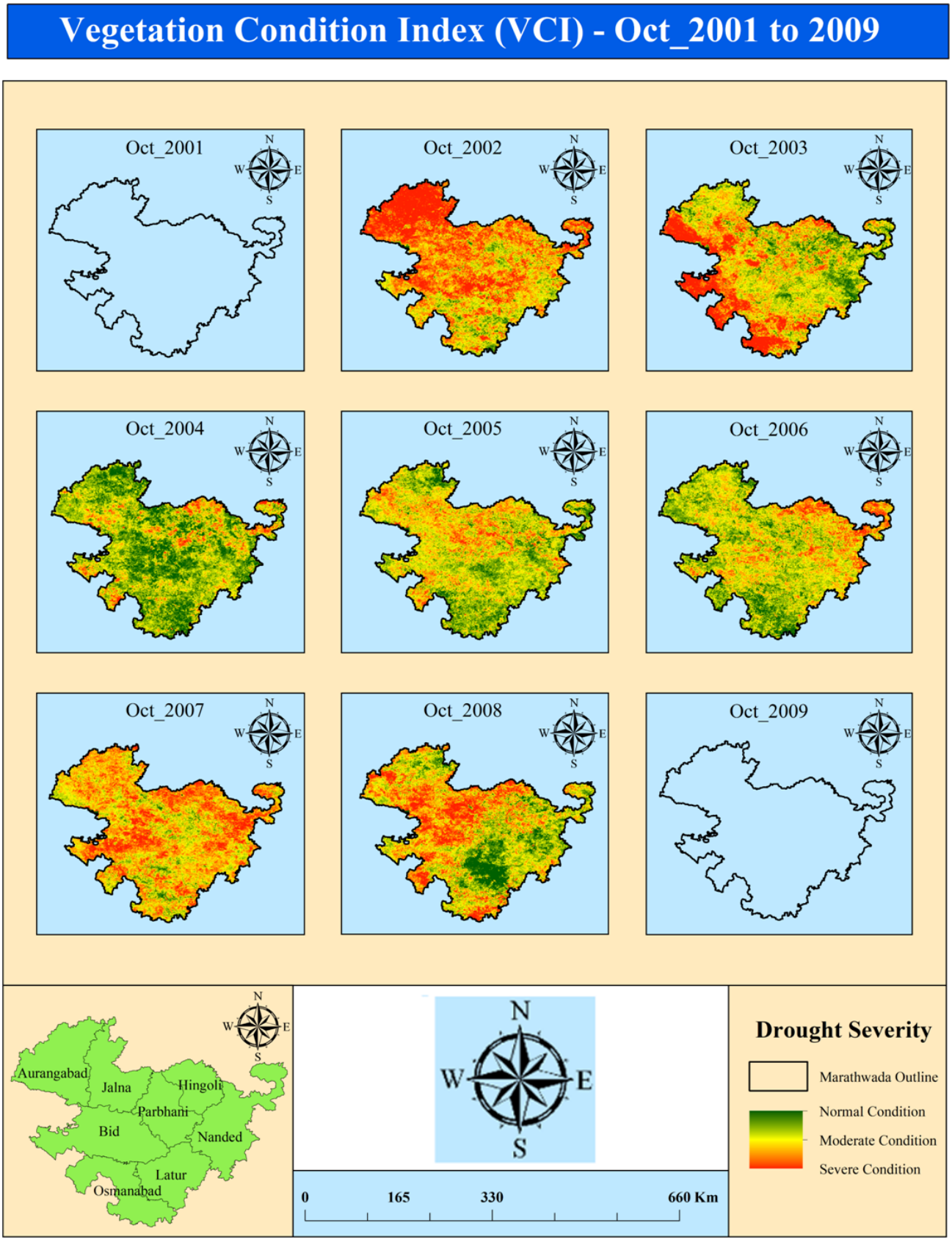


Figure 10. Vegetation Condition Index (VCI) - October 2001 to 2009

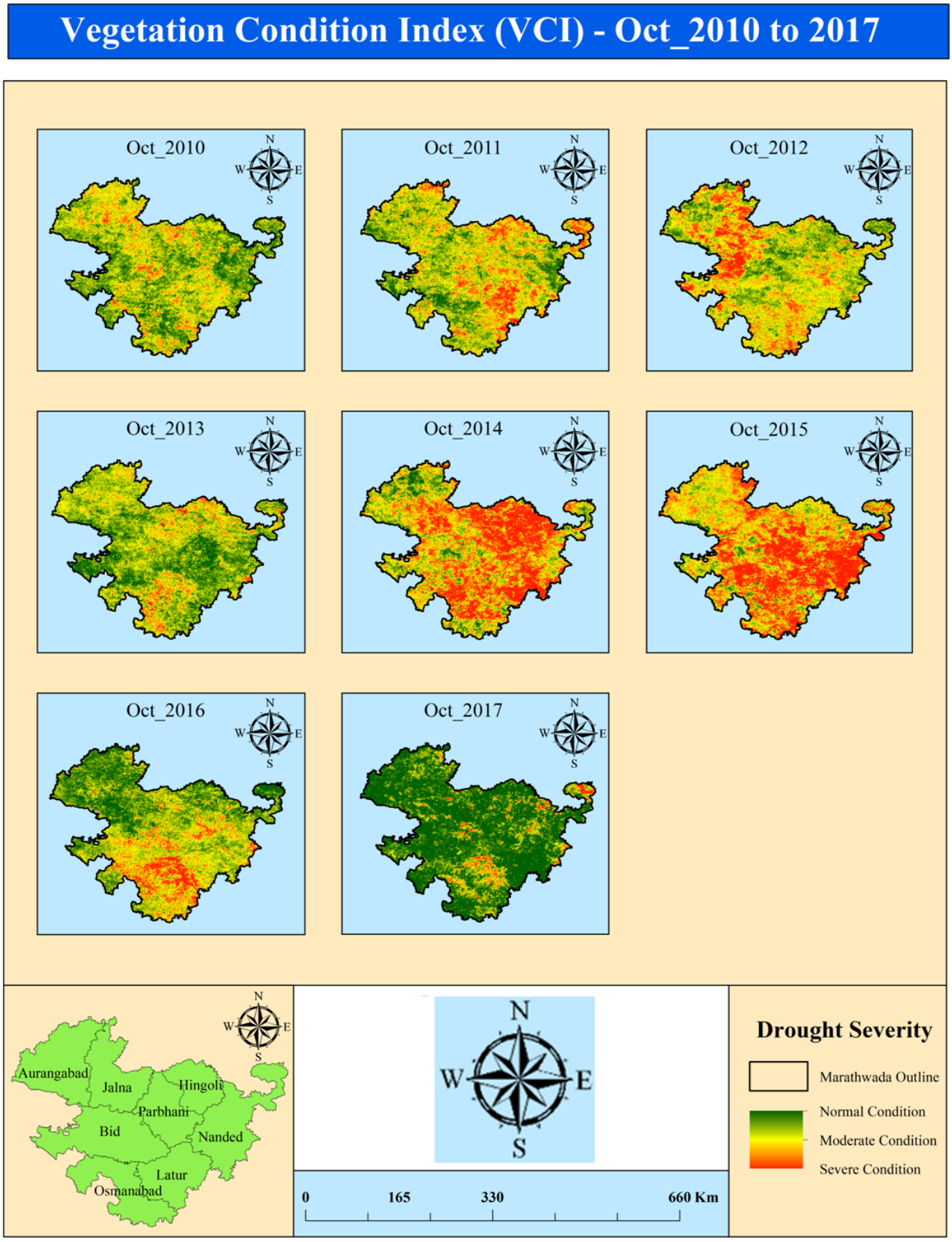


Figure 11. Vegetation Condition Index (VCI) - October 2010 to 2017

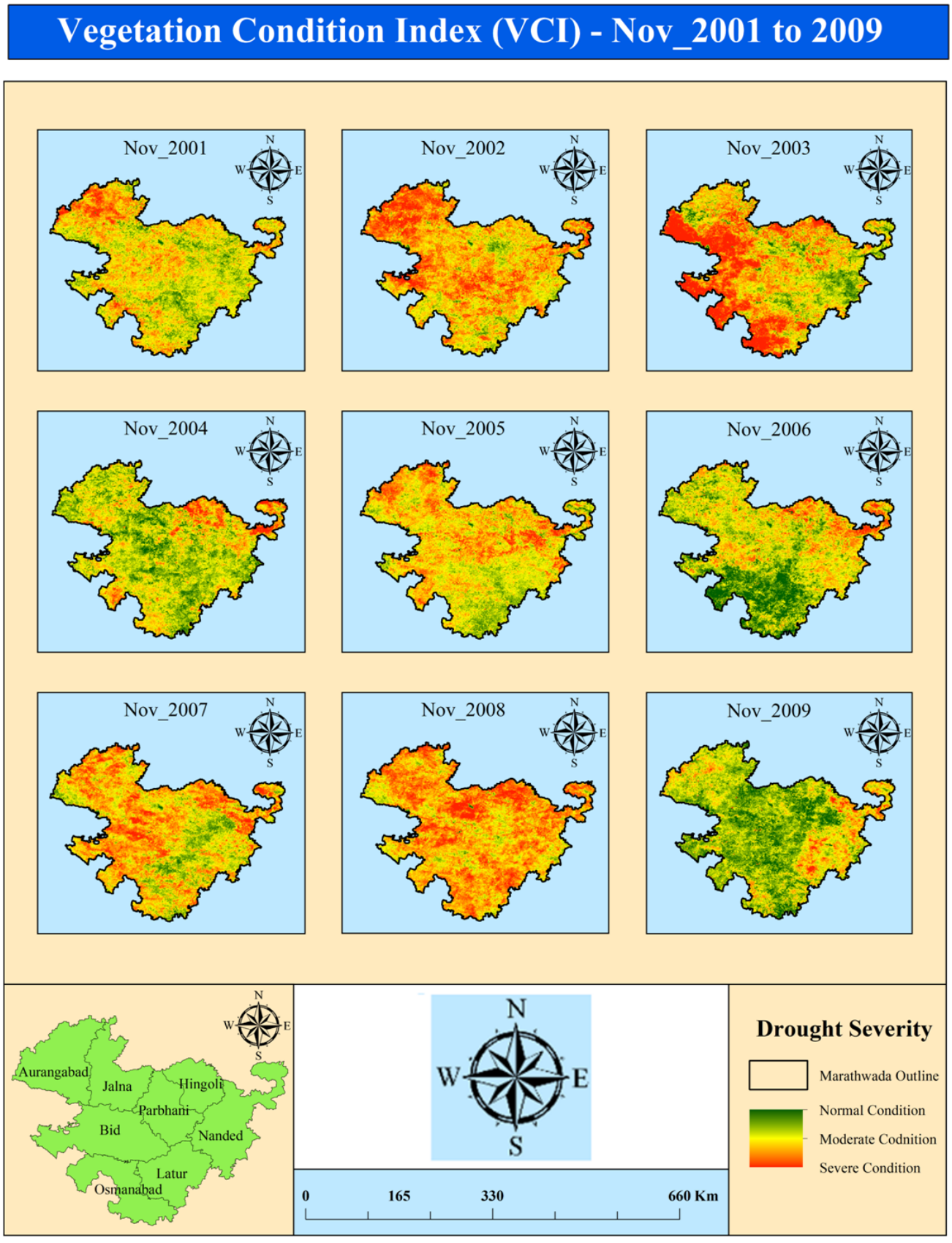


Figure 12. Vegetation Condition Index (VCI) - November 2001 to 2009

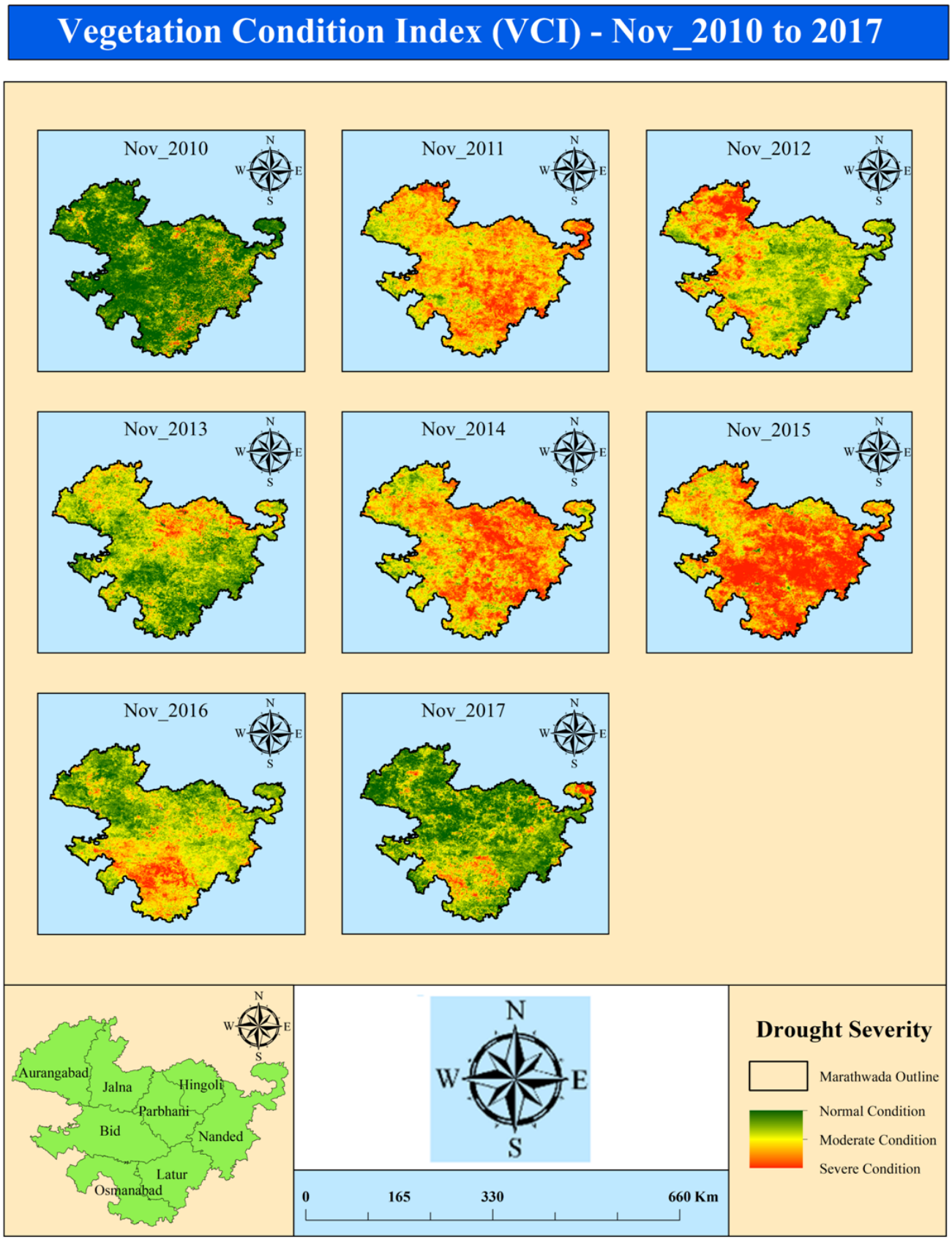


Figure 13. Vegetation Condition Index (VCI) - November 2010 to 2017

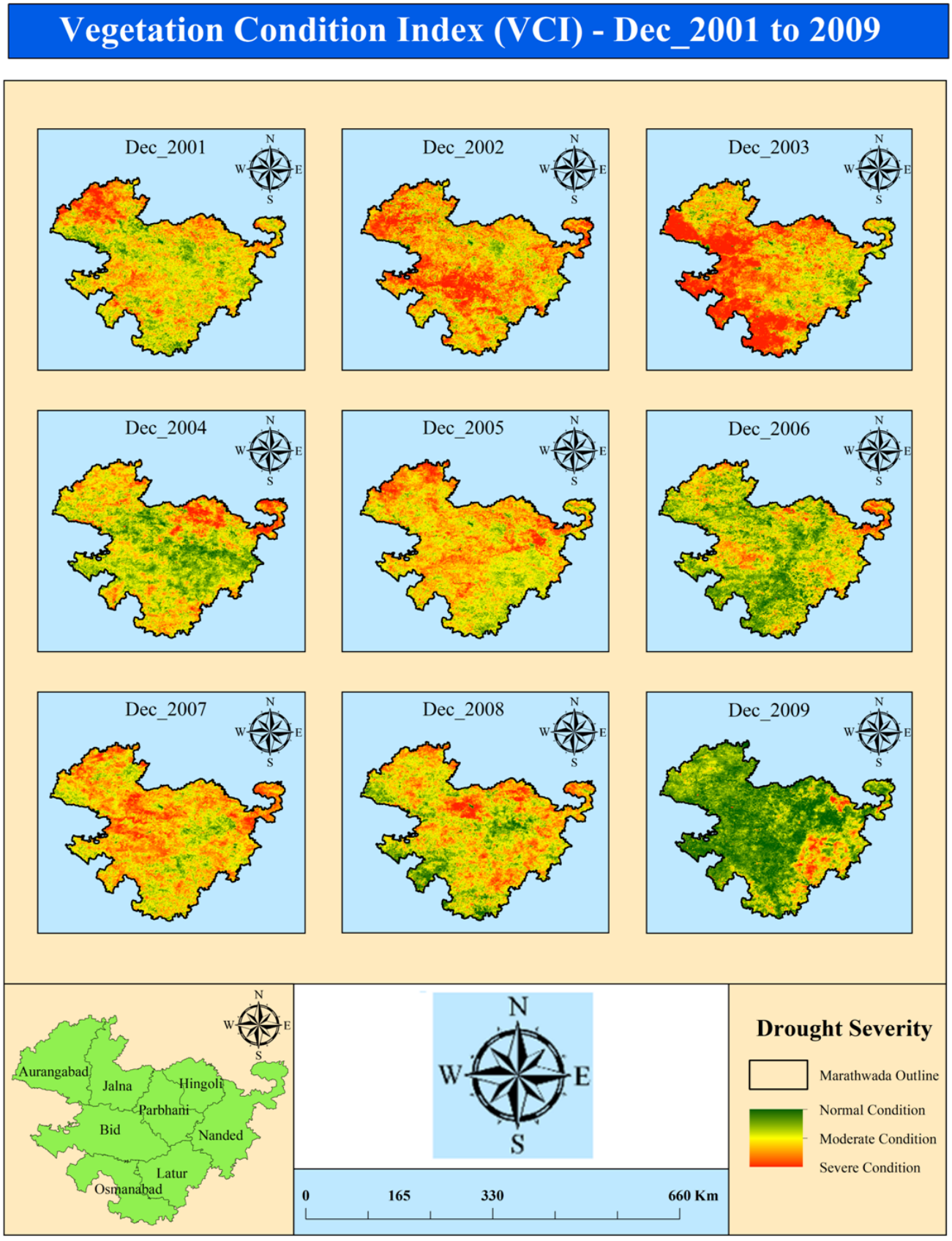


Figure 14. Vegetation Condition Index (VCI) - December 2001 to 2009

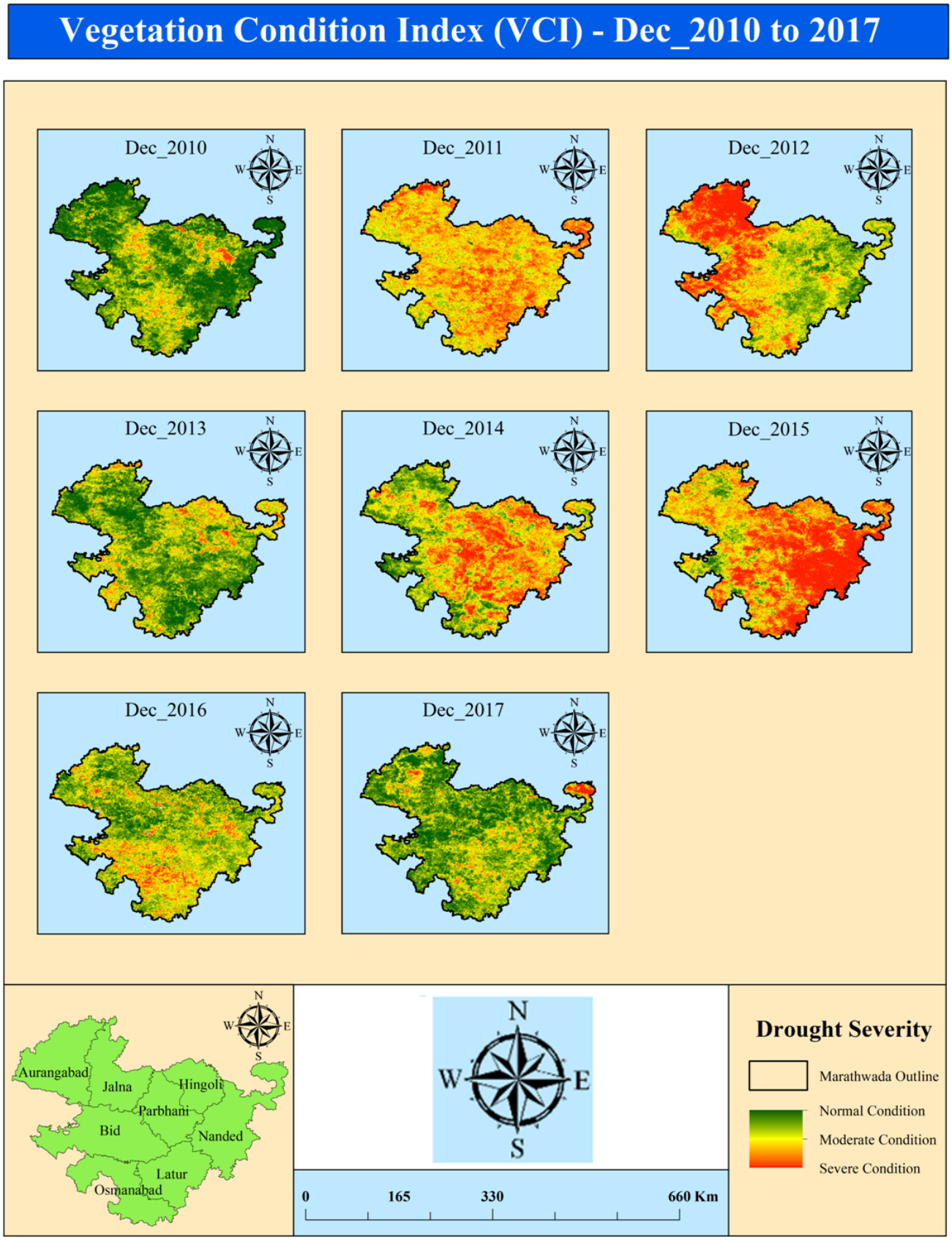


Figure 15. Vegetation Condition Index (VCI) - December 2010 to 2017

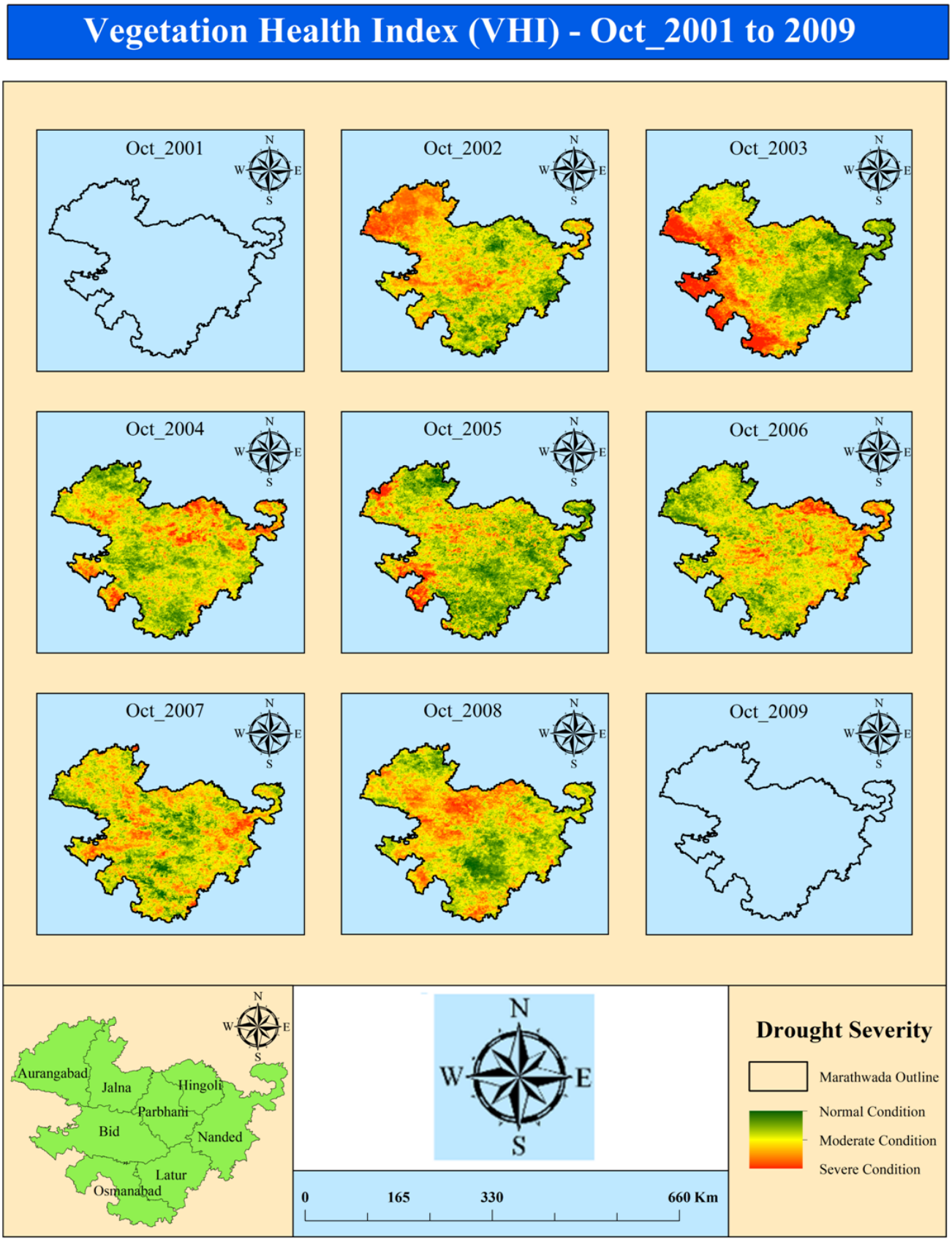


Figure 16. Vegetation Health Index (VHI) - October 2001 to 2009

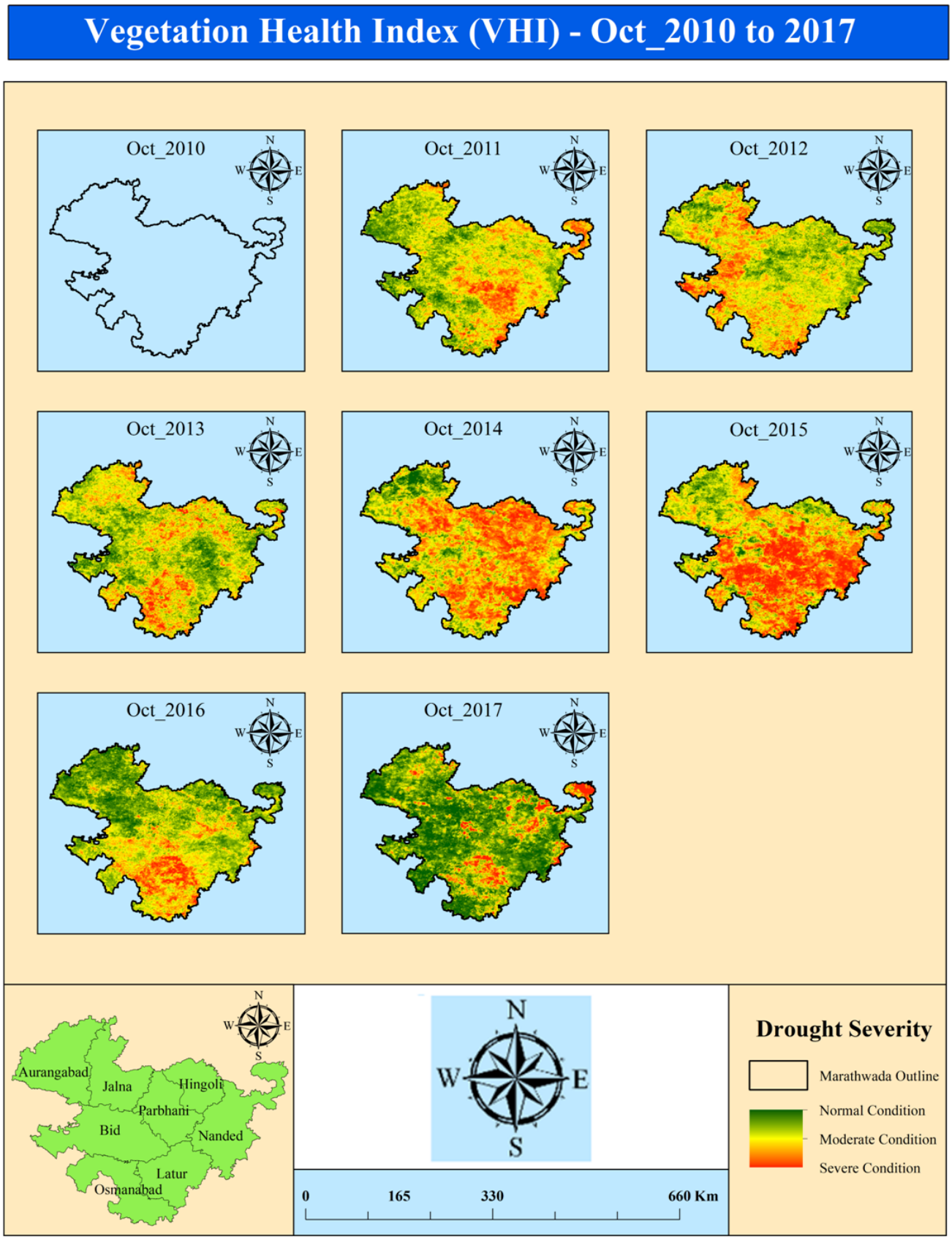


Figure 17. Vegetation Health Index (VHI) - October 2010 to 2017

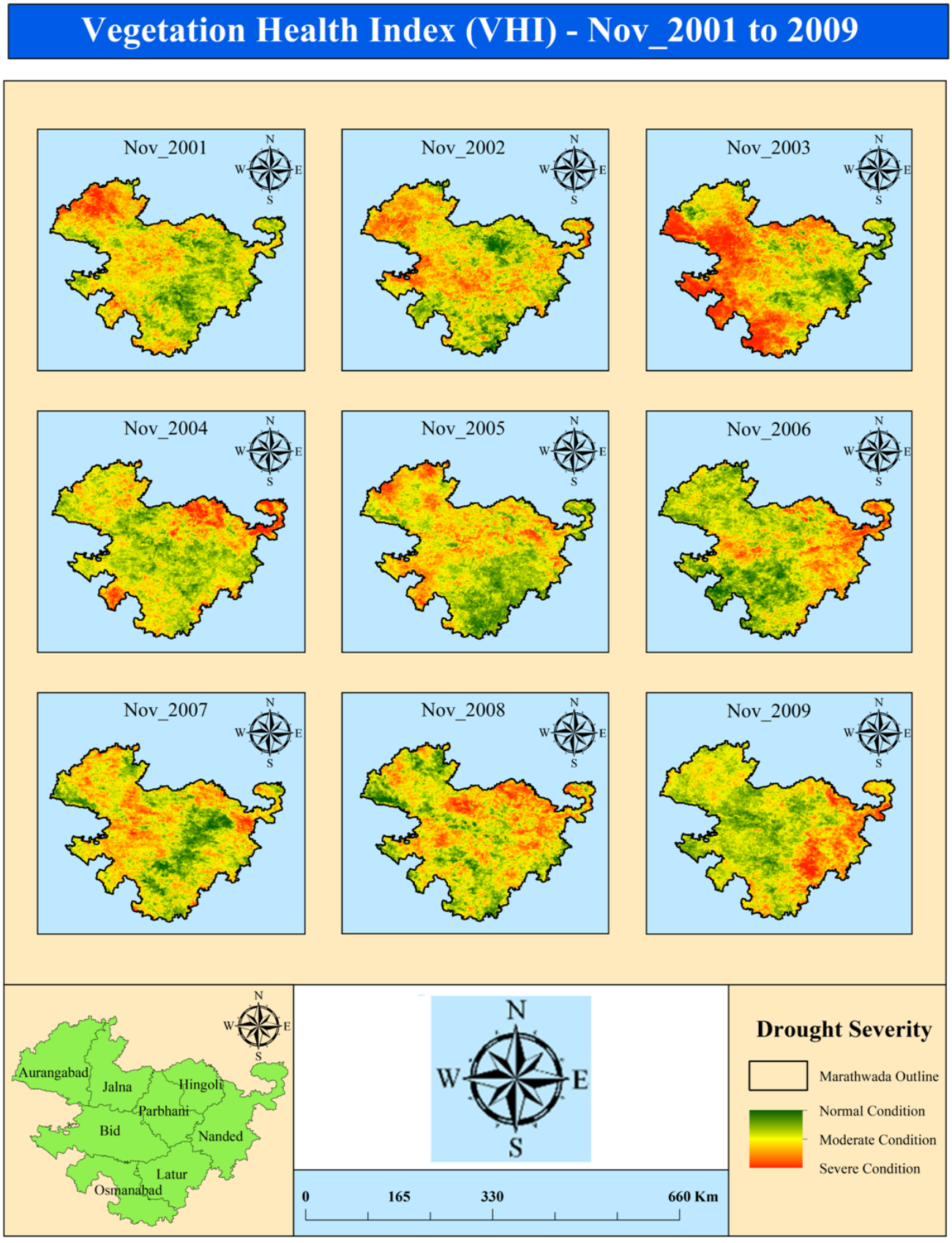


Figure 18. Vegetation Health Index (VHI) - November 2001 to 2009

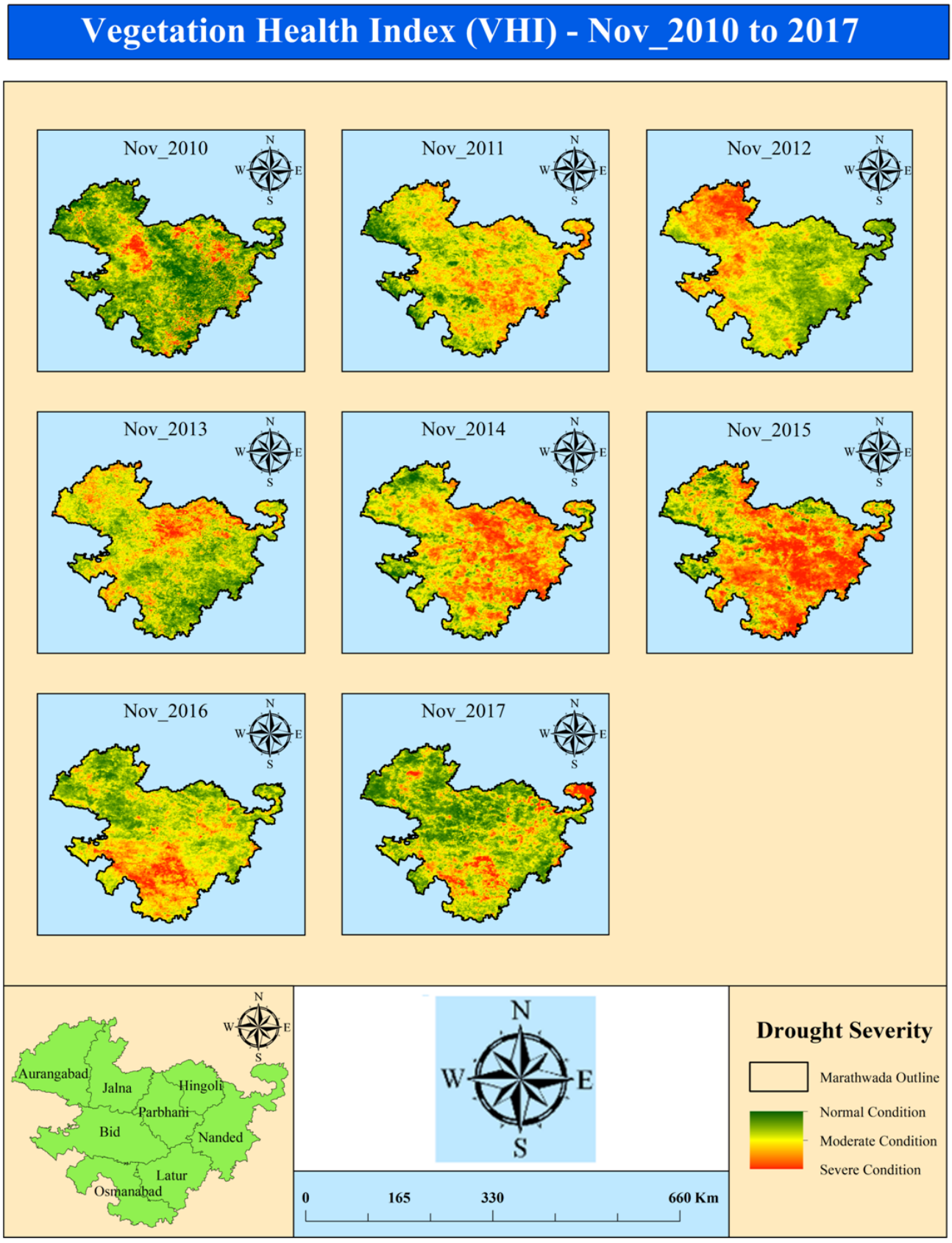


Figure 19. Vegetation Health Index (VHI) - November 2010 to 2017

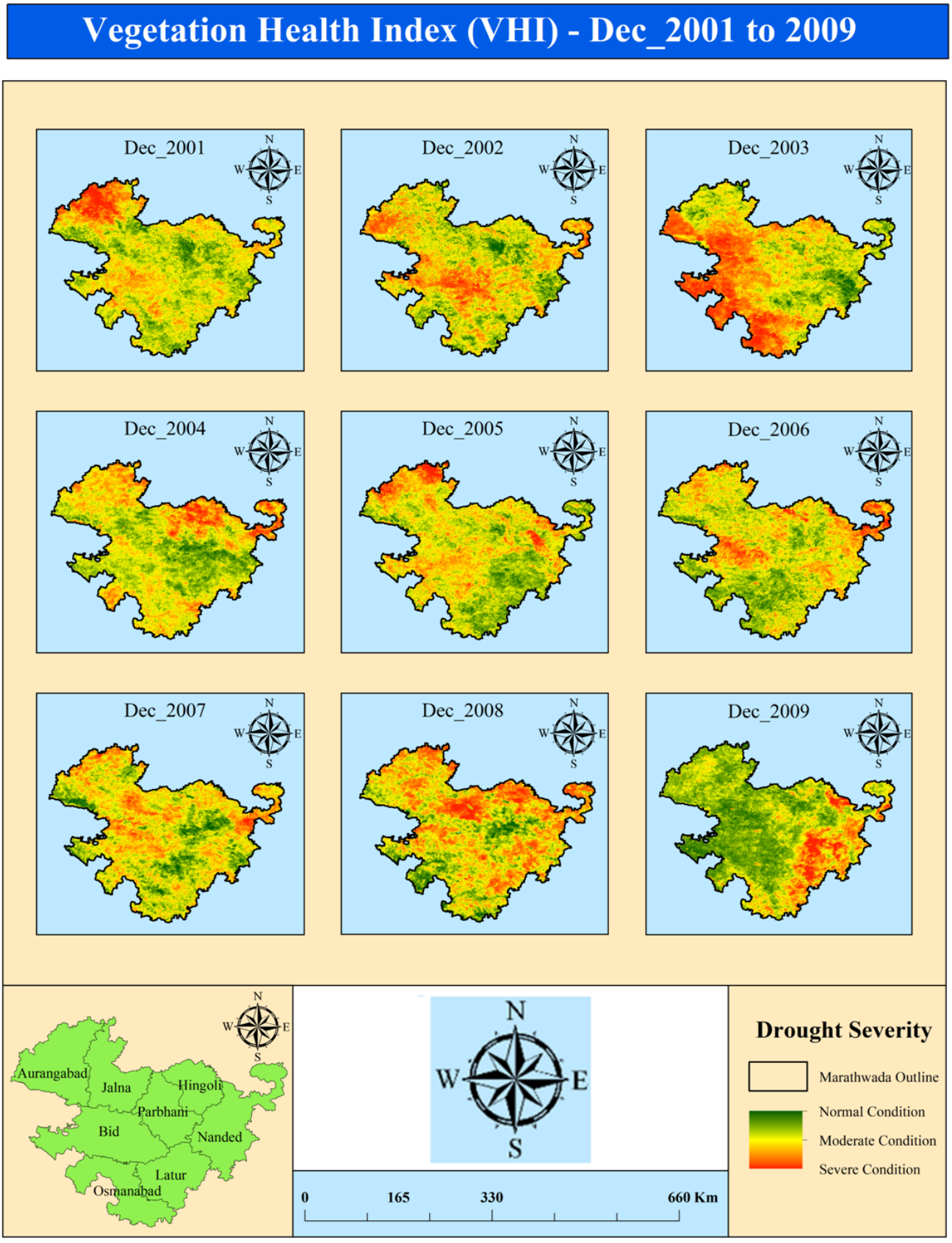


Figure 20. Vegetation Health Index (VHI) - December 2001 to 2009

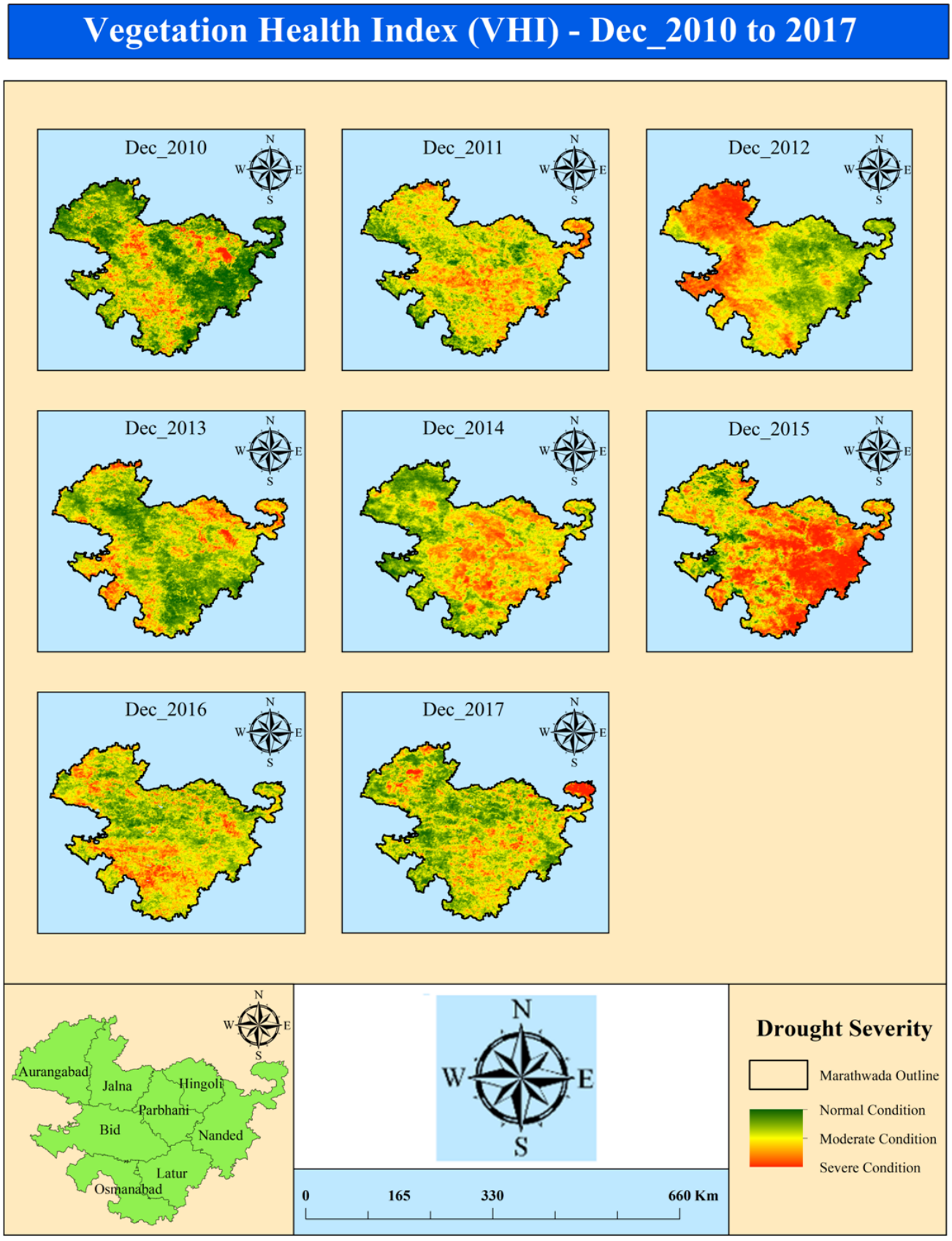


Figure 21. Vegetation Health Index (VHI) - December 2010 to 2017

In the present study agricultural drought-based remote sensing indices namely TCI, VCI and VHI were implemented for the year 2001 to 2017. The highly variable nature of rainfall over Marathwada can be understood from Figure 3, which is the primary reason for high drought frequency in the region. Scanty rainfall in the monsoon season over the Marathwada region leads to dryness and poor vegetation condition. Changes in the drought condition due to thermal impact, vegetation condition, and overall crop health were identified in this research. Overall crop health helps to understand the intensity of drought. As drought depends on various factors, such as rainfall deficiency, the moisture-holding capacity of the soil, heat waves, etc. These indices assist in identifying and assessing drought situations without any subjectivity.

TCI, VCI, and VHI results were obtained for the post-monsoon season (i.e. October, November, and December). The spatial pattern of the above three indices for the month of October is shown in Figure 4, Figure 5, Figure 10, Figure 11, Figure 16, and Figure 17. Wherever VHI shows green shades in these maps, it implies better NDVI and LST values over the same due to normal precipitation/moisture availability.

It can be noticed that overall the years 2014 and 2015 (Figure 5, Figure 11, and Figure 17) suffered a lot from dryness and thermal stress than any other years over Marathwada. Whereas if observed district-wise then, the western part of Beed and Aurangabad suffered from temperature stress. In the years 2007, 2008, and 2009 (Figure 4), most of the parts of Jalna and Hingoli suffered from moderate stress. In October 2001 (Figure 4), a severe drought situation occurred due to thermal stress covering almost the entire region. Due to the unavailability of October 2010 LST data, TCI for the same has not been generated (Figure 4).

In the case of VCI, October data for the years 2001 and 2009 were not available (Figure 10), and therefore, maps for these could not be generated. Like TCI, in VCI again 2014 and 2015 show poor vegetation conditions with severe drought (Figure 11). The years 2002, 2003, 2007, 2008, 2011, and 2012 appeared as moderate to poor vegetation conditions. If observed district-wise, Aurangabad suffered from dryness and less moisture during 2002 and 2012 identified with low NDVI values during the study. Good vegetation conditions with high NDVI values were observed in the years 2010, 2016, and 2017 where rainfall seems to be more prominent and sufficiently high (Figure 10, Figure 11). Except for the years 2014 and 2015, all the other years detected moderate to no drought conditions due to vegetation cover (moisture condition). In most of the years, moderate vegetation condition has been observed over the central part of Marathwada during the study of VCI.

It has been noticed that vegetation health is a combination of TCI and VCI representing thermal stress and unfavourable moisture condition. Therefore the results observed through VHI closely match with those of TCI and VCI results.

The TCI map for the month of November (Figure 6, Figure 7) revealed severe drought conditions during the years 2001, 2002, 2003, 2008, 2012, 2014, 2015 in some districts or over the entire region. Moderate to fair conditions were observed for the rest of the years. The three years 2003, 2014, and 2015 stand out with sweeping drought situations over the region. Similar conditions are represented by the VCI index represented in Figure 12 and Figure 13. But the year 2007 indicates more severity in the case of VCI than TCI. VHI results for the month of November brought out similar characteristics (Figure 18, Figure 19).

During December the severity of drought increased during the years 2008 and 2012 according to TCI (Figure 8, Figure 9). Severe conditions observed during October and November in the year 2015 are also observed in December, suggesting a prolonged drought situation during the end of the season. The same result can be concluded for the results obtained with VCI (Figure 14, Figure 15). The VHI map for the year 2012 (Figure 21) indicated more severe drought in the eastern part of the region (Aurangabad, Jalna, and Bid districts) compared to the months of October and November. Close observation during this year suggests that drought propagated from west to east in the region.

6. Conclusion

Over Marathwada, rainfall is very scanty which has led to semi-arid to arid climatic conditions. One of the geographical factors that contribute to the scanty rainfall is the location of this region in the rain shadow zone. Despite its climatic adversity, Marathwada contributes to about 10.10% of the state gross domestic product of Maharashtra, and 73.83% of its population is engaged in agriculture [16]. The study of drought over this region using remote-sensing-based indices has led to the identification of geographical variations in the intensity and severity of drought over this area. Since the remote sensing data are more continuous spatially, they are best suited to bring out the minor details of a hazard that inflicts a vast region. The present study has been successful in identifying the districts in the region which are more prone to drought. These are mostly the western districts of Aurangabad, Jalna, Bid, and Osmanabad. They are very well represented by the drought of 2003. But the drought in 2014 and 2015 also revealed that the eastern districts of Nanded, Parbhani, Latur, and to some extent Hingoli are equally susceptible to drought. In most case, it is noticed that drought usually begins first in the northeastern or eastern part. During the entire study period it was noticed that during more than 50% of years, at least half of the area was under moderate to severe drought conditions. With changing climate Marathwada has to be ready for higher rainfall variability and increased uncertainty of assured rainfall. The districts that are highly prone to drought will have to formulate plans for mitigation of future droughts.

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